

PI3K α inhibition blocks osteochondroprogenitor specification and the hyper-inflammatory response to prevent heterotopic ossification

Reviewed Preprint

v2 • July 2, 2024

Revised by authors

Reviewed Preprint

v1 • October 9, 2023

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Abstract

Heterotopic ossification (HO) occurs following mechanical trauma and burns, or congenitally in patients suffering from fibrodysplasia ossificans progressiva (FOP). Recently, we demonstrated that inhibitors of phosphatidylinositol 3-kinase α (PI3K α) may be a useful therapy for patients undergoing HO. In this study, using the already marketed BYL719/Alpelisib/Piqray drug, we have further confirmed these results, detailed the underlying mechanisms of action, and optimized the timing of the administration of BYL719. We found that BYL719 effectively prevents HO even when administered up to three to seven days after injury. We demonstrate in cell cultures and in a mouse model of HO that the major actions of BYL719 are on-target effects through the inhibition of PI3K α , without directly affecting ACVR1 or FOP-inducing ACVR1^{R206H} kinase activities. *In vivo*, we found that a lack of PI3K α in progenitors at injury sites is sufficient to prevent HO. Moreover, time course assays in HO lesions demonstrate that BYL719 not only blocks osteochondroprogenitor specification, but also reduces the inflammatory response. BYL719 inhibits the migration, proliferation and expression of pro-inflammatory cytokines in monocytes and mast cells, suggesting that BYL719 hampers the hyper-inflammatory status of HO lesions. Altogether, these results highlight the potential of PI3K α inhibition as a safe and effective therapeutic strategy for HO.

eLife assessment

This study provides **valuable** insights by demonstrating that BYL719 is a promising therapeutic agent for the treatment of heterotopic ossification (HO), with inhibition of PI3Ka via BYL719 appearing to be a critical factor. However, the results of the study are **incomplete** because BYL719 affects multiple intracellular signaling pathways beyond PI3Ka, and it thus remains uncertain whether BYL719 attenuates HO exclusively through suppression of the PI3Ka pathway or through modulation of alternative signaling pathways. A detailed elucidation of the molecular mechanisms of action of BYL719 is essential for a thorough understanding of its effects.

<https://doi.org/10.7554/eLife.91779.2.sa2>

Introduction

Heterotopic ossification (HO) is a disorder characterized by ectopic bone formation at extraskeletal sites, including skeletal muscle and connective tissues. Trauma-induced HO develops as a common post-operative complication after orthopedic surgeries (e.g., hip arthroplasty), blast injuries, deep burns and nervous system injuries (Hwang et al., 2022 [↗](#)). In addition to trauma-induced HO, fibrodysplasia ossificans progressiva (FOP) is a rare congenital autosomal dominant disorder also involving HO (Bravenboer et al., 2015 [↗](#)). Ectopic bones form progressively through endochondral ossification, mostly in episodic flare-ups associated with inflammation (Towler and Shore, 2022 [↗](#)). These cumulative osteogenic events lead to reduced mobility resulting from ankylosing joints. Affected individuals have a shorter life-span, most commonly due to thoracic insufficiency syndrome (Kaplan et al., 2010 [↗](#); Pignolo et al., 2016 [↗](#)).

HO reflects a shift from a normal tissue repair process to an aberrant reactivation of bone-forming developmental programs that require acute inflammation and the excessive expansion of local progenitors, followed by inappropriate differentiation of these progenitors into chondroblasts and osteoblasts which finally results in bone formation (Hwang et al., 2022 [↗](#)). In both trauma-induced HO and FOP, pathology appears following the excessive activation of receptors sensitive to Transforming growth factor- β (TGF β) superfamily members (Lees-Shepard et al., 2018 [↗](#); Wang et al., 2018 [↗](#)). FOP arises from gain-of-function mutations in the bone morphogenetic protein (BMP) type I receptor *ACVR1*, with the most common mutation being c.617G>A, R206H (Shore et al., 2006 [↗](#)). Higher SMAD1/5 mediated signaling of mutated *ACVR1* has been partially attributed to a loss of auto-inhibition of the receptor and mild hypersensitivity to BMP ligands (Chaikuad et al., 2012 [↗](#); Groppe et al., 2011 [↗](#); Van Dinther et al., 2010 [↗](#)). Signaling by mutated *ACVR1* still relies on ligand-induced heterotetrameric clustering but, unlike wild type *ACVR1*, does not require *ACVR2A/B* kinase activity (Agnew et al., 2021 [↗](#); Ramachandran et al., 2021 [↗](#)). More importantly, mutated *ACVR1* receptors alter their signaling specificity, abnormally activating SMAD1/5 and altering the non-canonical BMP signals (e.g., p38 or PI3K) in response to activin A (Hatsell et al., 2015 [↗](#); Hino et al., 2015 [↗](#); Valer et al., 2019a [↗](#)). Accordingly, evidence supports that activin A is necessary and sufficient for HO in FOP (Hatsell et al., 2015 [↗](#); Lees-Shepard et al., 2018 [↗](#); Upadhyay et al., 2017 [↗](#)). In contrast, activin A does not drive post-traumatic HO (Hwang et al., 2020 [↗](#)), and non-genetically-driven HO mostly arises from excessive BMP and TGF β signaling, with functional redundancy between different type I receptors of the TGF β -superfamily (Agarwal et al., 2017 [↗](#); Patel et al., 2022 [↗](#); Sorkin et al., 2020 [↗](#); Wang et al., 2018 [↗](#)).

Evidence points to mesenchymal fibroadipogenic precursors (FAPs), which are widely distributed in muscle and other connective tissues, as the key cell-of-origin that aberrantly undergo chondrogenesis and further ectopic bone formation (Dey et al., 2016; Eisner et al., 2020; Lees-Shepard et al., 2018). However, both FOP or non-genetic HO also require local tissue destruction and inflammation, which indicate that FAPs should be invariably primed for ectopic bone formation by this inflammatory microenvironment (Barruet et al., 2018; Convente et al., 2018; Hwang et al., 2022; Matsuo et al., 2019; Pignolo et al., 2013; Sorkin et al., 2020).

Inflammation in HO is characterized by an initial acute response following injury with the activation of innate immunity and the influx of neutrophils and monocytes (Convente et al., 2018; Hwang et al., 2022). Additionally, B and T cells of the adaptive system are also recruited (Chakkalakal et al., 2012; Convente et al., 2018). However, mice lacking B or T lymphocytes exhibited no delay in the development of heterotopic ossification (HO) after injury, indicating that these cells may play a subsequent role in the dissemination of bone lesions (Kan et al., 2009). During intermediate and late inflammatory stages, the recruitment of monocytes, macrophages and mast cells occurs, which in turn exerts autocrine and paracrine effects on nearby FAPs (Chakkalakal et al., 2012; Convente et al., 2018; Sorkin et al., 2020; Tu et al., 2022). Highlighting their relevance in HO, the depletion of mast cells and macrophages profoundly impairs genetic and non-genetic HO (Convente et al., 2018; Torossian et al., 2017; Tu et al., 2022). BMP receptors are robustly expressed in monocytes and macrophages and the expression of *ACVR1^{R206H}* has been shown to extend inflammatory responses in patient-derived macrophages (Barruet et al., 2018; Matsuo et al., 2021, 2019). Among the plethora of cytokines secreted by monocytes, macrophages and mast cells, both activin A and TGF- β stand out as extremely relevant for HO (Alessi Wolken et al., 2017; Hattell et al., 2015; Lees-Shepard et al., 2018; Patel et al., 2022; Sorkin et al., 2020; Upadhyay et al., 2017).

Genetic and pharmacological studies have indicated that osteochondroprogenitor specification and maturation depend on phosphatidylinositol 3-kinase- α (PI3K α) (Ford-Hutchinson et al., 2007; Fujita et al., 2004; Gámez et al., 2016; Ikegami et al., 2011). We found that PI3K signaling was also linked to HO, since inhibitors of PI3K α (BYL719/Alpelisib/Piqray) prevented HO in mouse models without major side-effects (Valer et al., 2019a). Mechanistically, PI3K α inhibitors hamper canonical and non-canonical BMP signaling, decreasing total and phosphorylated SMAD1/5 levels and reducing transcriptional responsiveness to BMPs/Activin A in mesenchymal progenitors (Gámez et al., 2016; Valer et al., 2019b). In the current study, we demonstrate that the pharmacological and genetic inhibition of PI3K α in HO progenitors at injury sites reduces HO *in vivo*. Moreover, envisioning a future translation into the clinic, we have optimized the administration of BYL719 and found that the delayed administration of BYL719, up to seven days after inflammatory injury, still prevents HO in mice. In addition, we found that BYL719 blocks osteochondroprogenitor specification and effectively reduces the essential inflammatory response. Altogether, the data presented here show the potent therapeutic effect of PI3K α inhibition on HO in mouse models.

Results

Delayed initiation of treatment with PI3K α inhibitor effectively prevents heterotopic ossification

Heterotopic ossification takes place following a precise temporal pattern of progenitor activation and the recruitment of distinct cell types at the ossification centers which causes early divergence from the normal skeletal muscle repair program. We previously found that the pharmacological administration of BYL719 prevents HO in a mouse model of HO (Valer et al., 2019a). Given that BYL719 is marketed for the treatment of cancer and overgrowth syndrome, it is a promising

therapeutic molecule in pathological HO. To further determine a potential therapeutic window for BYL719 and better understand the cellular targets of BYL719, we evaluated the effect of BYL719 when administered intermittently and several days after HO induction. For this, we used a conditional mouse model (*ACVR1^{Q207Dfl/fl}*) in which we combine the overexpression of Cre recombinase through adenoviral particles with the injection of cardiotoxin intramuscularly in the hindlimb (Fukuda et al., 2006 [↗](#)). In this model, we administered intermittently BYL719 (i.p. 25 mg/kg) or vehicle control starting one, three or seven days after HO induction. We also implemented a different administration regimen of BYL719 only for the initial three days following the injury (**Figure 1A** [↗](#)). None of the treatments led to significant changes in mouse body weight (Figure 1-Figure Supplement 1). HO volume was analyzed by micro computed tomography (μ CT) twenty-three days post-injury. The early administration of the treatment with BYL719 at day one or three after the onset of HO resulted in significantly lower HO; late administration seven days after the induction of HO was also partially effective (**Figure 1B** [↗](#) and **C** [↗](#)). Discontinuation of the treatment after three days did not prevent HO at day 23. These results indicate that the inhibition of PI3K α by BYL719 after injury prevents HO and this protection is still effective if the treatment is started several days after injury, expanding its therapeutic window.

Deficiency of PI3K α at injury sites is sufficient to partially prevent heterotopic ossification

Since other small molecules designed to target PI3K α have reported partial off-target effects on kinases other than PI3K α (Furet et al., 2013 [↗](#); Jamieson et al., 2011 [↗](#)), we aimed to confirm that the results obtained with BYL719 were due to the specific inhibition of PI3K α . Therefore, we developed a conditional mouse model (*ACVR1^{Q207Dfl/fl};p110 α ^{fl/fl}*) in which the Cre recombinase simultaneously drives the expression of *ACVR1^{Q207D}* and the deletion of the catalytic subunit of PI3K α (*p110 α*) upon intramuscular hindlimb combined injection of adenovirus-Cre and cardiotoxin to trigger HO. With this approach, Cre mediates the expression of *ACVR1^{Q207D}* and the deletion of PI3K α in the same cells, whereas cells recruited afterwards will remain wild type for both genes. As expected, μ CT analysis performed 23 days after injury showed extensive HO in mice expressing *ACVR1^{Q207D}*, which was prevented by intermittent treatment with BYL719 initiated one day after injury (**Figure 2A**, **2B** [↗](#) and **2D** [↗](#)). Genetic deletion of *p110 α* in mice expressing *ACVR1^{Q207D}* led to a reduction in HO, which was further enhanced by BYL719 (**Figure 2B** [↗](#)). Importantly, no significant changes in the weight of the mice were observed, confirming the absence of severe toxicity (Figure 2-Figure Supplement 1A). Histomorphometrically, *ACVR1^{Q207D}* mice wild type for PI3K α developed islands and/or bone spurs with a well-organized bone structure. However, in PI3K α deficient mice *ACVR1^{Q207D}* expression only led to minor ectopic calcifications that were already surrounded by fully regenerated muscle tissue on the 23rd day after injury (**Figure 2D** [↗](#), Figure 2-Figure Supplement 1B, Figure 2-Figure Supplement 2). Accordingly, analysis of the ratio bone volume/tissue volume (BV/TV) and the correlation between bone volume and BV/TV of individual mice was clearly different between *p110 α* genotypes, irrespective of BYL719 treatment (**Figure 2C** [↗](#), Figure 2-Figure Supplement 1C). These results demonstrate that the abrogation of PI3K α activity (either genetic or pharmacologically) in cells at injury sites is relevant for their inhibitory effects in HO. This might be attributed to the direct effects of BYL719 on progenitor cell expansion and chondroblast specification, as well as its possible ability to reduce the inflammatory response required for HO.

Genetic or pharmacological inhibition of PI3K α prevents osteochondroprogenitor specification

Next, we aimed to assess the effects of PI3K α inhibition in osteochondroprogenitor cells. For this, we isolated bone marrow-derived mesenchymal stem cells (BM-MSCs) from *p110 α ^{fl/fl}* mice and transduced them *ex-vivo* with retrovirus expressing *Acvr1* (*wild type* and *R206H*) with or without

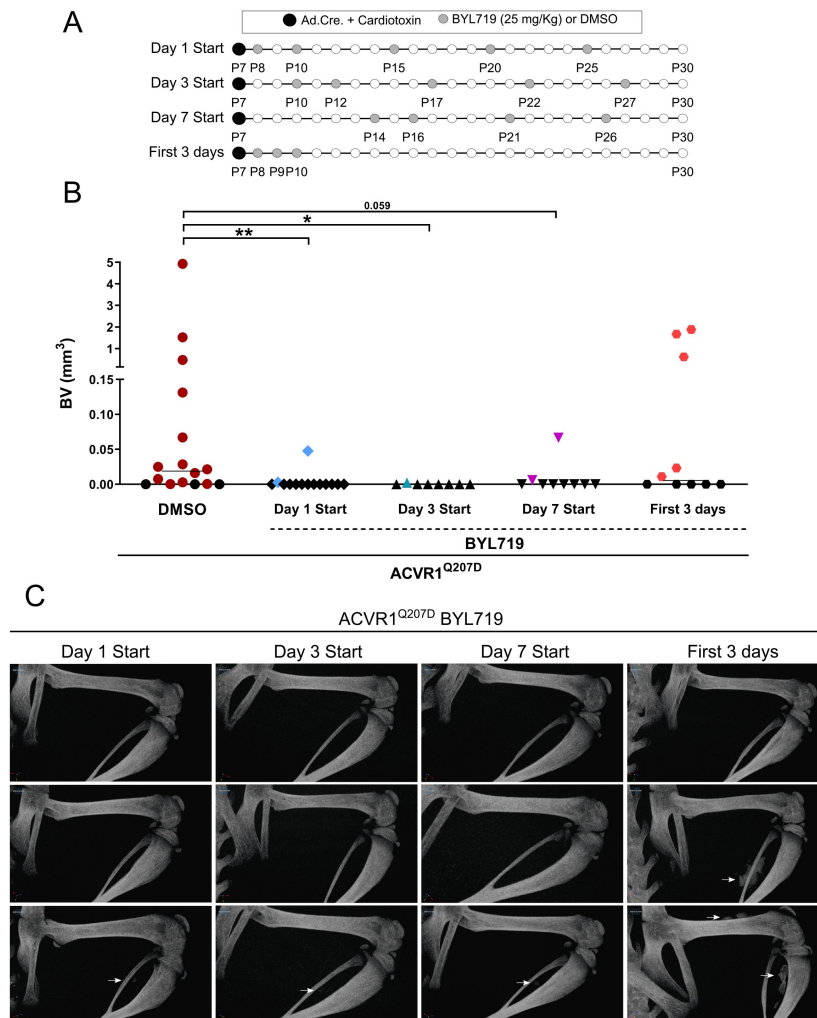


Figure 1.

(A) Heterotopic ossification was induced at P7 through the injection of adenovirus-Cre (Ad.Cre) and cardiotoxin in the mice hindlimb. The pattern of 5 administrations of DMSO or BYL719 (25 mg/kg) is indicated by grey dots started at P8 (start day 1), P10 (delayed start day 3), and P14 (delayed start day 7). One group was administered DMSO or BYL719 (25 mg/kg) only at P8, P9, and P10 (first 3 days post-HO-induction), as indicated by grey dots. The final time-point, P30, was conserved between experimental groups. (B) Quantification of heterotopic ossifications bone volume (BV) (mm³) of each experimental group. Colored symbols indicate the presence of heterotopic ossifications. Black symbols indicate the absence of heterotopic ossifications. Individual mouse values with group median are shown. * $P < 0.05$, ** $P < 0.01$, Kruskal-Wallis test with Dunn's multiple comparison test. (C) Representative 3D microtomography images of the injected hindlimbs of 3 different mice for each experimental group. White arrows point to heterotopic ossification.

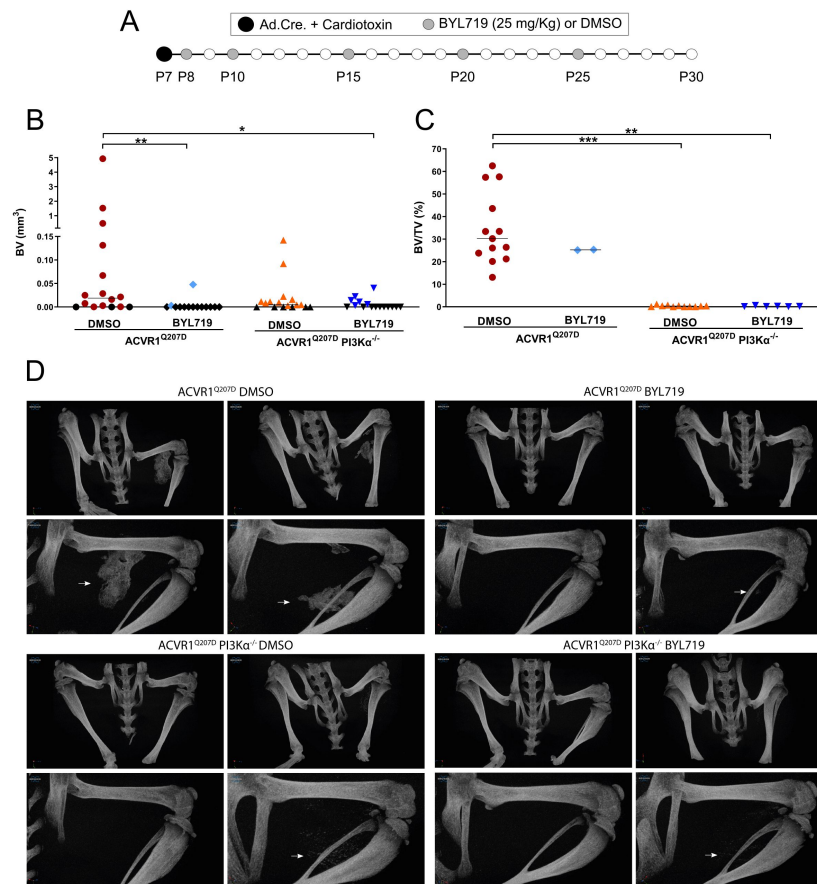


Figure 2.

(A) Heterotopic ossification was induced at P7 through the injection of Adenovirus-Cre (Ad.Cre) and cardiotoxin in the mice hindlimb. Either DMSO (vehicle) or BYL719 (25 mg/kg) were injected following the scheme indicated with grey dots, starting at P8 (B) Quantification of heterotopic ossifications bone volume (BV)(mm³) of each experimental group. Colored symbols indicate the presence of heterotopic ossifications. Black symbols indicate the absence of heterotopic ossifications. Individual mouse values with group median are shown. *P<0.05, **P<0.01, Kruskal-Wallis test with Dunn's multiple comparison test. (C) Quantification of the ratio of bone volume per tissue volume (bone volume/tissue volume, BV/TV) within heterotopic ossifications of each experimental group, including only mice with detected HO. Data shown are of each individual mouse with the group median. **P<0.01, ***P<0.001, Kruskal-Wallis test with Dunn's multiple comparison test. (D) Representative 3D frontal microtomography images of the injected hindlimbs of mice for each experimental group and a detailed close-up image for each selected mouse. White arrows show heterotopic ossification in the close-up images.

Cre recombinase. The gene expression levels of exogenous mRNA of *Acvr1* and *Acvr1*^{R206H} was similar (**Figure 3A**). As anticipated, transduction with Cre viruses significantly reduced the expression levels of p110α to approximately 40%. This is consistent with an estimated transduction efficiency of nearly 60% (**Figure 3B**). The overexpression of *Acvr1*^{R206H} increased basal and activin-dependent expression of canonical (*Id1* and *Sp7*) and activin-dependent expression of non-canonical (*Ptgs2*) BMP target genes (**Figure 3C**), known markers and drivers of osteochondrogenic differentiation (Nakashima et al., 2002; Wang et al., 2013). Both pharmacological and genetic inhibition of PI3Kα reduced canonical and non-canonical ACVR1^{R206H} transcriptional responses induced by activin A. Of note, stimulation with activin A increased its own transcription (*Inhba*) in *Acvr1*^{R206H}-transduced BM-MSCs, which was also inhibited by BYL719 (**Figure 3C**). Therefore, BYL719 may also prevent osteochondroprogenitor differentiation indirectly, via transcriptional inhibition of *INHBA*/Activin A expression. These results demonstrate a direct effect of BYL719 on osteochondroprogenitor cells and suggest that the effects of BYL719 on ACVR1-downstream signaling are mainly on target, due to PI3Kα kinase inhibition.

It is known that 2-aminothiazole-derived PI3Kα inhibitors could also inhibit ACVR1 kinase activity (Jamieson et al., 2011). Therefore, we investigated whether BYL719 effects could be explained by the reduction in ACVR1 kinase activity. For this, we made use of a kinase target engagement approach. Upon transient transfection of Nanoluciferase-tagged TGF-β receptors, we incubated the cells with BYL719 and/or an ATP-like tracer analogue. In the absence of an ATP competitor molecule, the tracer analogue and Nanoluciferase enzyme stay in close proximity, thereby allowing for bioluminescence resonance energy transfer (BRET) between the Nanoluciferase (energy donor, emitting at 460 nm) and the tracer analogue (energy acceptor) Once excited, the acceptor molecule emits fluorescence (610 nm) (**Figure 4A** and **B**). BYL719 incubated at 1 and 10 μM failed to significantly inhibit the acceptor emission of the BMP type I receptors ACVR11, ACVR1, BMPR1A, ACVR1B, TGFBR1 and the mutant receptor ACVR1^{R206H} (**Figure 4C**), without significant changes in the protein levels of each receptor (Figure 4-Figure Supplement 1). In addition, we demonstrated that BYL719 does not target the activity of the type II kinase receptors ACVR2A, ACVR2B, BMPR2 and TGFBR2. We confirmed these results using *in vitro* kinase activity assays with recombinant ACVR1^{R206H}. BYL719 exhibited no major effects on ACVR1^{R206H} activity up to 10 μM (**Figure 4D**).

Given that BYL719 does not directly target the activity of ACVR1 or related receptors, we established an *in vitro* model using human MSCs (hMSCs) stably overexpressing ACVR1 or ACVR1^{R206H} to profile further the main targeted pathways. In this model, recombinant Activin A was able to induce SMAD1/5 activation and the expression of ACVR1 downstream targets *ID1* and *ID3* in cells over-expressing ACVR1^{R206H} (Figure 5-Figure Supplement 1A and B). We performed micromass chondrogenic differentiation assays in differentiation media supplemented with TGFβ1, with or without activin A and BYL719 in parental cells and cells transduced with ACVR1^{WT} or ACVR1^{R206H}. Coincubation with BYL719 (10 μM) completely inhibited the formation of a chondrogenic glycosaminoglycans-rich matrix in response to TGFβ1 plus activin A in the different cell types (**Figure 5A** and **B**) and partially abrogated the expression of the chondrogenic genes *ACAN* and *MMP13* (**Figure 5C**). These results confirmed that BYL719 could inhibit chondrogenic differentiation of parental hMSCs and also the chondrogenic specification irrespective of the expression of either wild type or mutant ACVR1. Furthermore, in these cells, coincubation with BYL719 (10 μM) was able to repress Activin A-induced alkaline phosphatase activity upon 7 and 11 days of culture, respectively (**Figure 5D** and **E**).

To further identify the effects of BYL719, we performed bulk RNA sequencing (GSE237512) where we preincubated hMSCs-ACVR1^{WT} or hMSCs-ACVR1^{R206H} with or without BYL719 for 30 minutes, which was followed by stimulation with two high affinity ligands for ACVR1, Activin A (50ng/ml) or BMP6 (50ng/ml) for 1 hour (**Figure 6A**). Consistent with our results on osteogenic progenitor cells, gene ontology analysis of differentially expressed genes between cells expressing ACVR1^{WT}

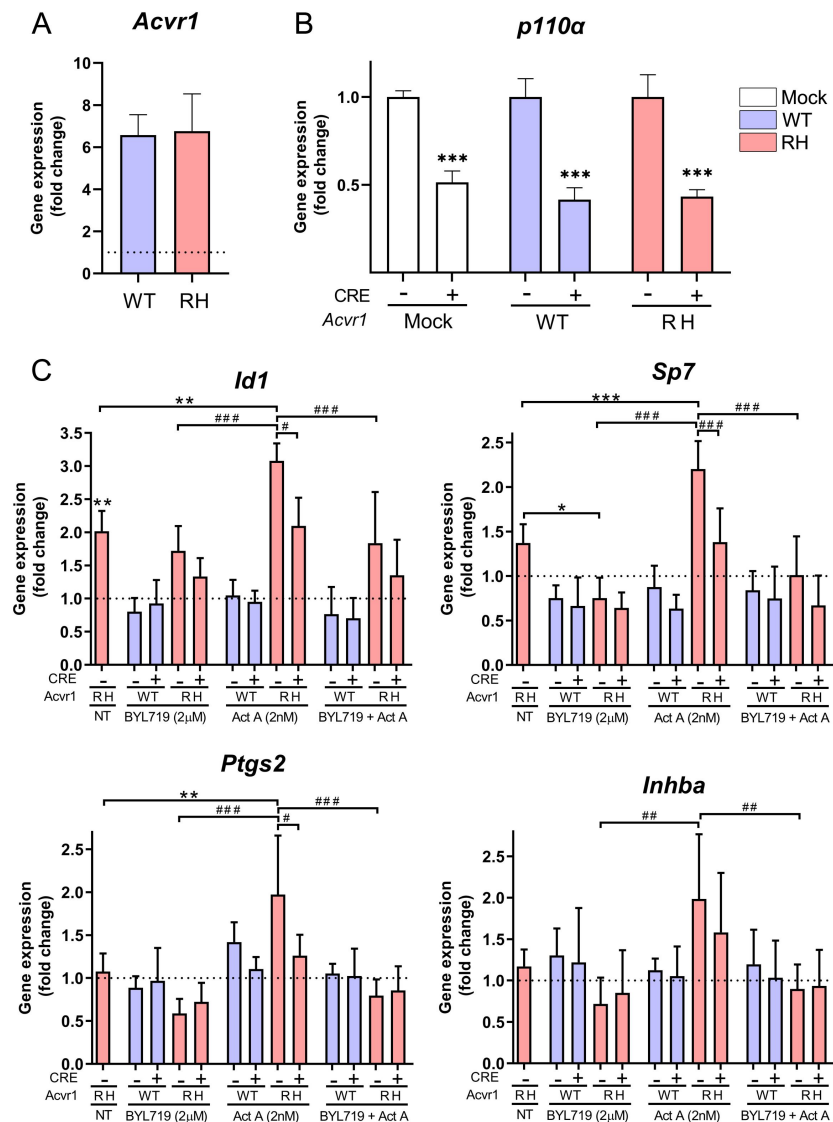


Figure 3.

Analysis of *Acvr1* gene expression by qPCR in BM-MSCs from *p110α^{fl/fl}* mice infected with virus expressing wild-type *Acvr1* (WT) or *Acvr1^{R206H}* (RH). (A) The endogenous expression level of *Acvr1* gene in mock-transfected BM-MSCs is shown as a dotted horizontal line. Data are shown as mean $\pm$ SD ($n = 12$ per group). Unpaired t-test between transfected groups. (B) Gene expression analysis of *p110α* in BM-MSCs from *p110α^{fl/fl}* mice, infected with virus expressing wild-type *Acvr1* (WT) or *Acvr1^{R206H}* (RH) and/or Cre co-infection. Data are shown as mean $\pm$ SD ($n = 3$ per group). *** $P < 0.001$, two-way ANOVA with Tukey's multiple comparisons test. (C) mRNA expression of canonical (*Id1* and *Sp7*), non-canonical (*Ptgs2*) target genes, and activin A (*Inhba*) in BM-MSCs *p110α^{fl/fl}* transfected with *Acvr1* (wild type or R206H) with or without Cre recombinase. Cells were NT (not treated), or treated with BYL719 (2μM) and/or activin A (2nM). Expression data was normalized to those of control cells which were transfected only with *Acvr1* WT without any treatment, shown as a dotted horizontal line. Asterisks (*) refer to the differences between different conditions of *Acvr1* RH cells compared to control cells. Hash signs (#) refer to the differences between different conditions of *Acvr1* RH cells compared to *Acvr1* RH cells without Cre recombinase and treated with activin A. Data are shown as mean $\pm$ SD ($n = 6$ per group). * or # $P < 0.05$, ** or ## $P < 0.01$, *** or ### $P < 0.001$, two-way ANOVA with Tukey's multiple comparisons test.

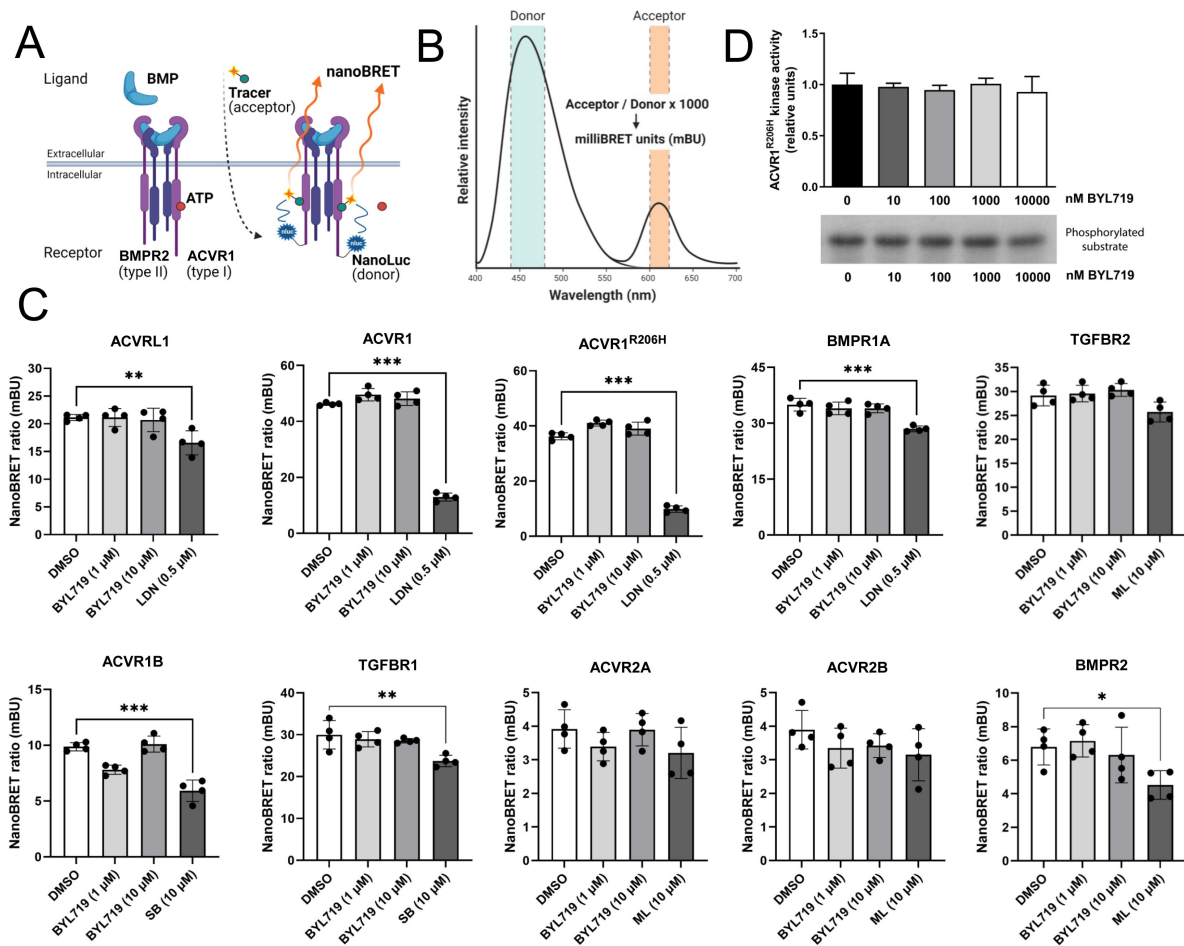


Figure 4.

(A) A schematic depiction of nano-bioluminescence resonance energy transfer (NanoBRET) target engagement assays. TGF- β receptors-Nanoluciferase fusion proteins are expressed in COS-1 cells and an ATP-like tracer analogue in close proximity with the Nanoluciferase donor allows energy transfer from the Nanoluciferase donor to the fluorescent tracer acceptor. An ATP analogue molecule (type I inhibitor) will compete with the fluorescent tracer, impairing close proximity donor-acceptor and reducing the nanoBRET ratio. (B) The nanoBRET emission spectra consist of the Nanoluciferase donor (460nm) and the fluorescent acceptor (610nm). The nanoBRET ratio is shown as milliBRET units (mBU) by dividing the acceptor emission by the donor emission times 1000. (C) NanoBRET target engagement analyses of ACVRL1, ACVR1, ACVR1^{R206H}, BMPR1A, ACVR1B, TGFB1, TGFB2, ACVR2A, ACVR2B and BMPR2 testing 1 or 10 μ M BYL719 with $n=4$. As controls, LDN193189 (0.5 μ M), SB431542 (10 μ M), and ML347 (10 μ M) were used. Data are shown as mean $\pm$ SD. One-way ANOVA with Dunnett's multiple comparisons test. (D) Casein phosphorylation by ACVR1^{R206H} kinase. Phosphorylation was performed in the presence of ACVR1^{R206H} kinase and increasing concentrations of the PI3K α inhibitor BYL719. Quantification of kinase activity. Data shown as mean $\pm$ SD ($n=4$ independent experiments). One-way ANOVA with Tukey's multiple comparisons test. * $P<0.05$, ** $P<0.01$, *** $P<0.001$.

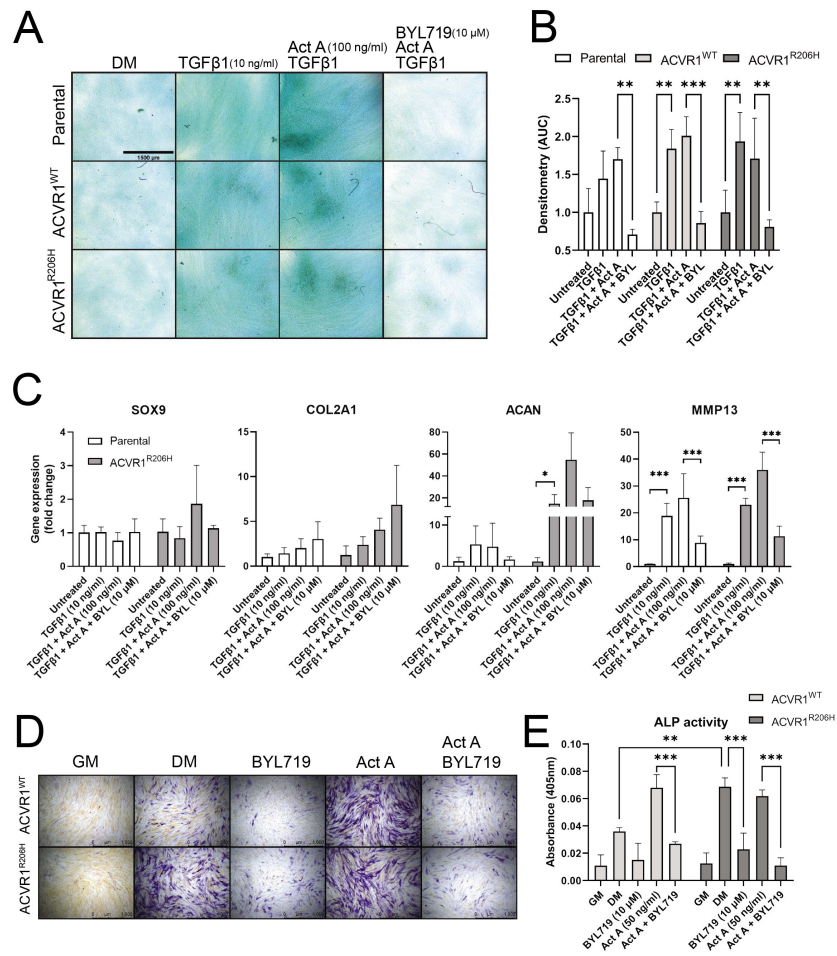


Figure 5.

Analyses of chondrogenic progenitor specification of parental human MSCs (hMSCs), hMSC-ACVR1^{WT} or hMSC-ACVR1^{R206H} treated with TGF-β1 (10 ng/ml) and Activin A (100 ng/ml), with and without BYL719 (10 μM) as indicated (A) Representative images of Alcian Blue staining of micromass cultures after 3 weeks of chondrogenic differentiation. Scale bar represents 1500 μm. (B) Densitometry analysis of the Alcian Blue staining in (A) (n=3 per group). (C) RT-qPCR results of *SOX9*, *COL2A1*, *ACAN* and *MMP13* after chondrogenic progenitor specification (n=4 per condition). (D) Representative images of the ALP staining after one week of chondrogenic progenitor specification. Scale bar represents 1000 μm. (E) Quantification of the ALP activity (n=3 per group). Data are shown as mean ± SD. In all figure panels, significance is detailed between the indicated conditions. * P<0.05, ** P<0.01, *** P<0.001, two-way ANOVA with Tukey's multiple comparisons test.

and ACVR1^{R206H} resulted in the robust regulation of relevant biological processes. Ossification (GO:0001503) and osteoblast differentiation (GO:0001649) were detected as two of the ten most significantly differentially regulated biological processes between these conditions (**Figure 6B** [↗](#)). Moreover, using gene ontology, we analyzed ossification and osteoblast differentiation biological processes in presence of ACVR1^{WT} or ACVR1^{R206H} receptor, with different ligands (BMP6 or Activin A), and with or without BYL719 inhibitor. The addition of BYL719 (1 μ M) resulted in a downregulation of these GO terms for all conditions except activin A stimulated ACVR1^{WT} cells (**Figure 6C** [↗](#)).

When comparing gene expressions profiles between ACVR1^{R206H} treated with Activin A, with or without BYL719, Gene Set Enrichment Analysis (GSEA) of our GO terms of interest showed negative normalized enrichment scores, consistent with our gene ontology analysis (**Figure 6D** [↗](#)). By looking at the leading edge we could investigate the most relevant genes within these GO terms, which included relevant genes of these pathways such as *PTGS2*, *MSX2*, *SOX9*, and *BMP2* (**Figure 6E** [↗](#)). Moreover, several inflammatory cytokine genes (e.g. *IL6* and *IL15*) and inflammatory signaling pathways (e.g. TNF and NF κ B), were downregulated by BYL719 (Figure 6-Figure Supplement 1A and B). Therefore, we hypothesized that, in addition to preventing the activation of osteogenic and chondrogenic differentiation pathways in progenitor cells, part of the mechanism by which BYL719 prevents HO *in vivo* may be due to the modulation of inflammatory responses.

PI3K α inhibition reduces proliferation, migration and inflammatory cytokine expression in monocytes, macrophages and mast cells

Next, we used several models of disease-relevant immune cells to study the effect of pharmacological PI3K α inhibition. Activation of ACVR1 signaling with BMP6 did not significantly modify the proliferation rate of any of the cell lines tested, that is, monocytes (THP1), macrophages (RAW264.7) and mast cells (HMC1) (**Figure 7A**, **B** [↗](#) and **C** [↗](#)). However, incubation with BYL719 *in vitro* strongly reduced their growth rate at 2 μ M and almost completely arrested their proliferation at 10 μ M. It is well known that monocytes are actively recruited from circulation to the ossification sites early after injury (Sorkin et al., 2020 [↗](#)). We also examined the effects of BYL719 in proliferation in additional cell types involved in HO, such as myoblast and mesenchymal progenitors. BYL719 reduced the proliferation of myoblast and mesenchymal progenitors *in vitro* (**Figure 7** [↗](#), Figure Supplement 1 A and B). However, the reduction in the proliferation did not reach the extent observed in monocytes or macrophages. We expanded our analysis by interrogating chemotactic migration of monocytes in transwell assays. Whereas monocytes actively migrated towards FBS as a chemotactic agent, the incubation of monocytes with BYL719 deeply diminished their migratory ability (**Figure 7D** [↗](#)). These results suggest that BYL719 is able to diminish monocyte recruitment, while reducing also the expansion of inflammatory cells at injury sites.

Upon tissue injury, a multitude of pro-inflammatory and anti-inflammatory cytokines are released at the site of injury. This triggers the recruitment and infiltration of various cell types, including immune cells and osteochondrogenic progenitors. We examined the expression of distinct cytokines upon ACVR1 stimulation with BMP6 in the presence of BYL719 in monocytic, macrophagic and mast cell lines. In monocytes, BYL719 inhibited the expression of the pro-inflammatory cytokine *TNF- α* and the expression of activin A (*INHBA*), whereas the expression of transforming growth factor *TGF- β 1* was increased by BMP6 and not modified by the addition BYL719 (**Figure 7E** [↗](#)). *CCL2* (MCP-1), a chemokine inducing the recruitment of monocytes and macrophages into inflamed tissues as well as the M2 polarization of macrophages (Sierra-Filardi et al., 2014 [↗](#)), was significantly induced by BMP6 and partially reduced by BYL719 (**Figure 7E** [↗](#)). In macrophages, BYL719 reduced the expression of pan-macrophage markers *Adgre1* (F4/80), *Cd68* and the M1 macrophage marker *Cd80* (**Figure 7F** [↗](#)). Conversely, BYL719 increased the expression

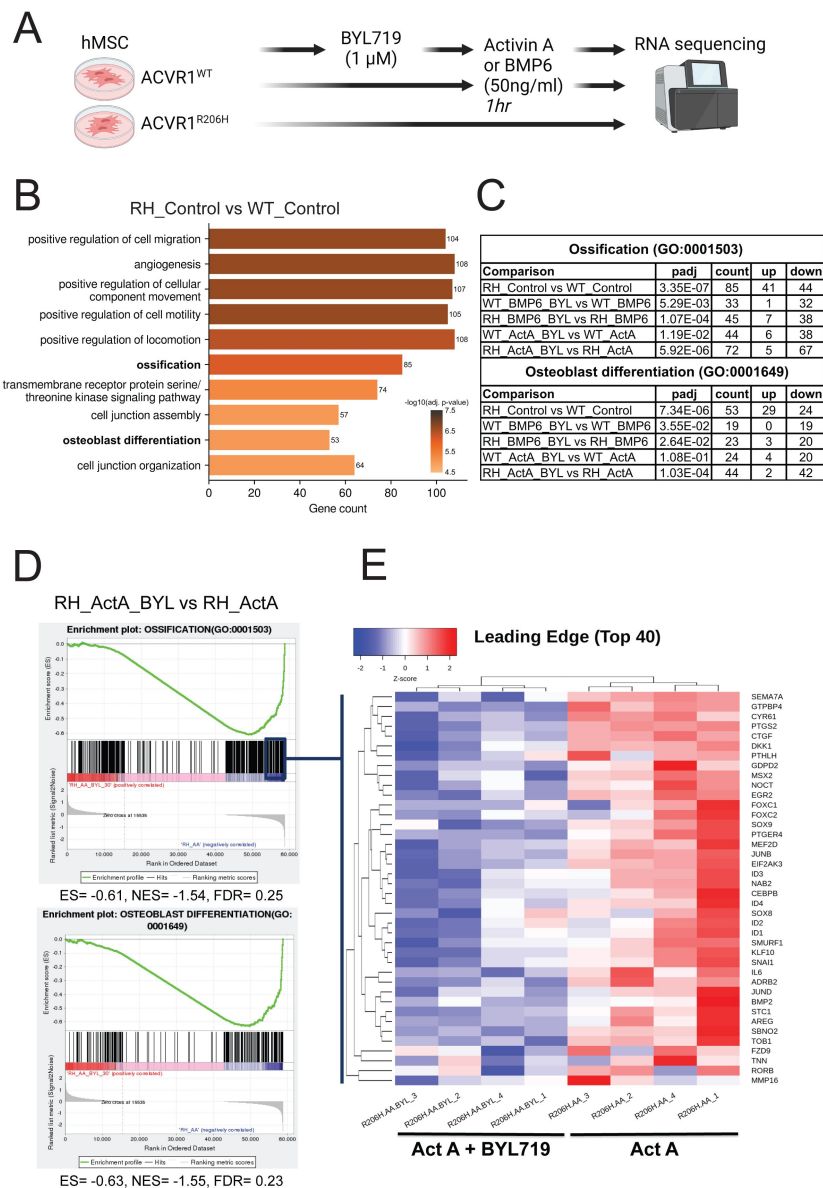


Figure 6.

(A) Schematic depiction of the experimental setup, involving bulk RNA sequencing of human MSC-ACVR1^{WT} and ACVR1^{R206H} upon overnight starvation, 30 minutes pre-treatment with or without 1 μ M BYL719 and with or without 1 hour Activin A (50 ng/ml) or BMP6 (50ng/ml) stimulation. (B) The top 10 most significant gene ontology (GO) terms using all (up- and down-regulated) differentially expressed genes between cells expressing ACVR1^{WT} and ACVR1^{R206H} under control conditions. Ossification (GO:0001503) and osteoblast differentiation (GO:0001649) were detected within the top 10 of the differentially regulated biological processes. (C) Table of GO terms ossification and osteoblast differentiation upon GO enrichment analysis in all tested conditions ACVR1^{WT} or ACVR1^{R206H} cells, stimulated with BMP6 or Activin A and treated with BYL719. Statistical significance upon GO enrichment analysis, count of total differentially expressed genes (DEGs), and their classification as up- or down-regulated DEGs are detailed for each comparison. (D) Gene Set Enrichment Analysis (GSEA) with the groups RH_ActA_BYL vs RH_ActA, showing the enrichment plots of the gene ontology sets ossification (GO: 0001503) and osteoblast differentiation (GO:0001649). (E) A heatmap of the top 40 most relevant genes within the ossification GO geneset derived from the leading-edge subset of the enrichment plot (n=4 per group). Sample names are detailed as receptor (R206H) ligand (AA, Act A, Activin A)_inhibitor (BYL719, if present)_replicate#.

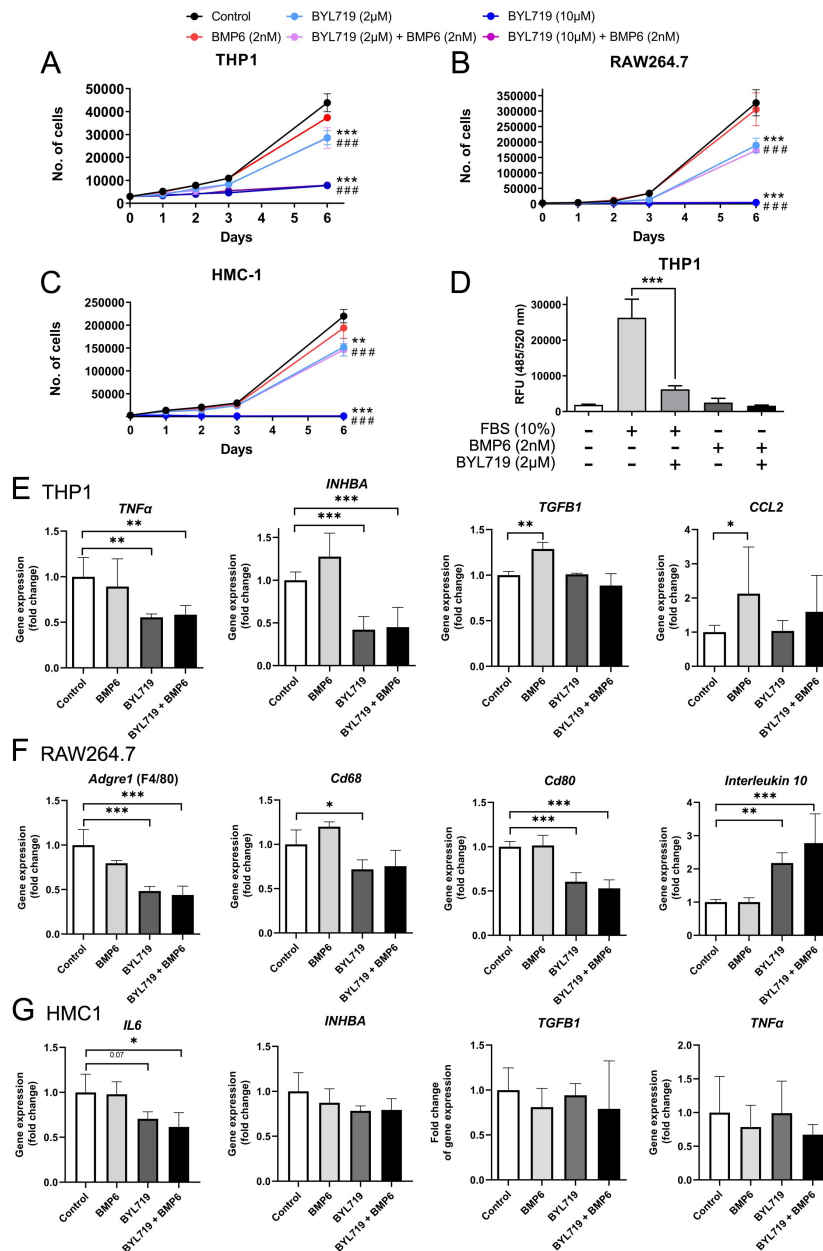


Figure 7.

(A, B, C) Proliferation assays of THP1 (A), RAW264.7 (B), and HMC-1 (C) cells. Cells were cultured for 6 days and in control conditions, with BMP6 (2nM) and/or BYL719 (2 or 10μM). Areas under the proliferation curves were compared by one-way ANOVA with Dunnett's multiple comparisons test against the control. (*) refers to significant differences between control and single treatments (BMP6 or BYL719). (#) refers to significant differences between control and combined treatments. Data shown as mean ± SD (n = 4 per group). ** or # P<0.01, *** or # P<0.001. (D) Migration assay of THP1 cells with control conditions, or with FBS or BMP6 as chemotactic agents with or without BYL719 treatment. Data are shown as mean ± SD (n = 4 per group). ***P<0.001, two-way ANOVA with Tukey's multiple comparisons test. (E, F, G) Gene expression assays of THP1 (E), RAW264.7 (F) and HMC-1 (G) cells. Cells were treated for 48 hours with BYL719 (2μM) and/or BMP6 (2nM). Data are shown as mean ± SD (n = 4 per group). *P<0.05, **P<0.01, ***P<0.001, one-way ANOVA with Dunnett's multiple comparisons test, significance shown between control group and other groups.

of the M2 macrophage marker *Il10*, regardless of the presence or absence of BMP6. In the human mast cell line HMC1, BYL719 inhibited the expression of the pro-inflammatory cytokine *IL6*, while the expression of *TGFβ1*, *TNFα*, and activin A (*INHBA*) remained unchanged upon the addition of either BMP6 or BYL719 (**Figure 7G**). We also analyzed the effects of BYL719 on myogenic differentiation *in vitro*. Addition of 2 or 10 μM BYL719 was able to abolish the expression of the myogenic markers *MyoD1* and myosin heavy-chain 1 (*Myh1*) while barely altering the expression of the myofibroblast marker α-SMA (*Acta2*) (Figure 7-Figure Supplement 1C). Similarly, BYL719 (at 2 or 10 μM) was able to block formation of myotubes after 7 days of myogenic differentiation *in vitro* (Figure 7-Figure Supplement 1D). Altogether, these data suggest that PI3Kα inhibition reduces the expression of certain pro-inflammatory cytokines and might skew macrophages towards M2. Moreover, even though these results should be further validated in cells carrying mutated ACVR1, the reduction on activin A expression by BYL719 observed in inflammatory cells and mesenchymal progenitors could be relevant for HO in FOP.

PI3K_α inhibition reduces the hyperinflammatory response in heterotopic ossification

To confirm whether BYL719 can modulate excessive inflammation at the injured sites, we examined its impact on monocytes, macrophages and mast cells *in vivo*. These cell populations are known to increase in number two to four days after injury (Convente et al., 2018; Sorkin et al., 2020), which coincides with the effects observed following early administration of BYL719. Hindlimb muscle samples were collected at two, four, nine, sixteen and twenty-three days post-induction of HO with Cre viruses and cardiotoxin in *ACVR1^{Q207Dfl/fl}* mice (**Figure 8A**). The group of treated animals received BYL719 intermittent treatment starting one day after injury. After sixteen days, structures of mineralized bone became detectable in vehicle-treated mice (Figure 8-Figure Supplement 1A and B). The number of F4/80-positive monocytes/macrophages was high two days after injury and remained elevated in the HO lesions throughout the ossification process (**Figure 8B** and **C**). BYL719-treated animals showed an almost 50% reduction in the number of F4/80 positive cells at day two post-injury onwards. Consistent with previous studies, we observed an increased number of mast cells at injury sites. Their numbers peaked at four days and only dropped twenty-three days after injury in vehicle-treated mice (**Figure 8D** and **E**). In BYL719-treated mice, mast cells were increased up to day 9, but their number was completely normalized to preinjury levels already at day 16 (**Figure 8D** and **E**). This observation is consistent with the complete muscle regeneration observed in BYL719-treated mice compared to untreated mice, which are actively developing ossifications at that time (Figure 8-Figure Supplement 2). These results confirm a functionally relevant dual role of PI3Kα inhibition, by preventing the differentiation of osteochondrogenic progenitor cells and also by reducing the local inflammatory response.

Discussion

We previously showed that pharmacological PI3Kα inhibition using BYL719 has the potential to suppress HO by increasing SMAD1/5 degradation, reducing transcriptional responsiveness to BMPs, and blocking non-canonical responses such as the activation of AKT/mTOR (Valer et al., 2019a). In this manuscript, we confirmed these results through both pharmacological and genetic approaches, and further expanded the insights into the molecular and cellular mechanisms responsible for the therapeutic effect of PI3K inhibition. Here, we optimized the therapeutic window for BYL719 administration and found that the delayed administration of BYL719 effectively prevents HO. Moreover, *in vivo* simultaneous genetic inhibition of *p110α* and the activation of mutated *ACVR1* at injury sites partially prevents HO, and we demonstrate that inhibition of PI3Kα inhibits osteochondroprogenitor specification. We determined that BYL719 does not inhibit recombinant ACVR1^{R206H} kinase activity *in vitro* and does not behave as an ATP

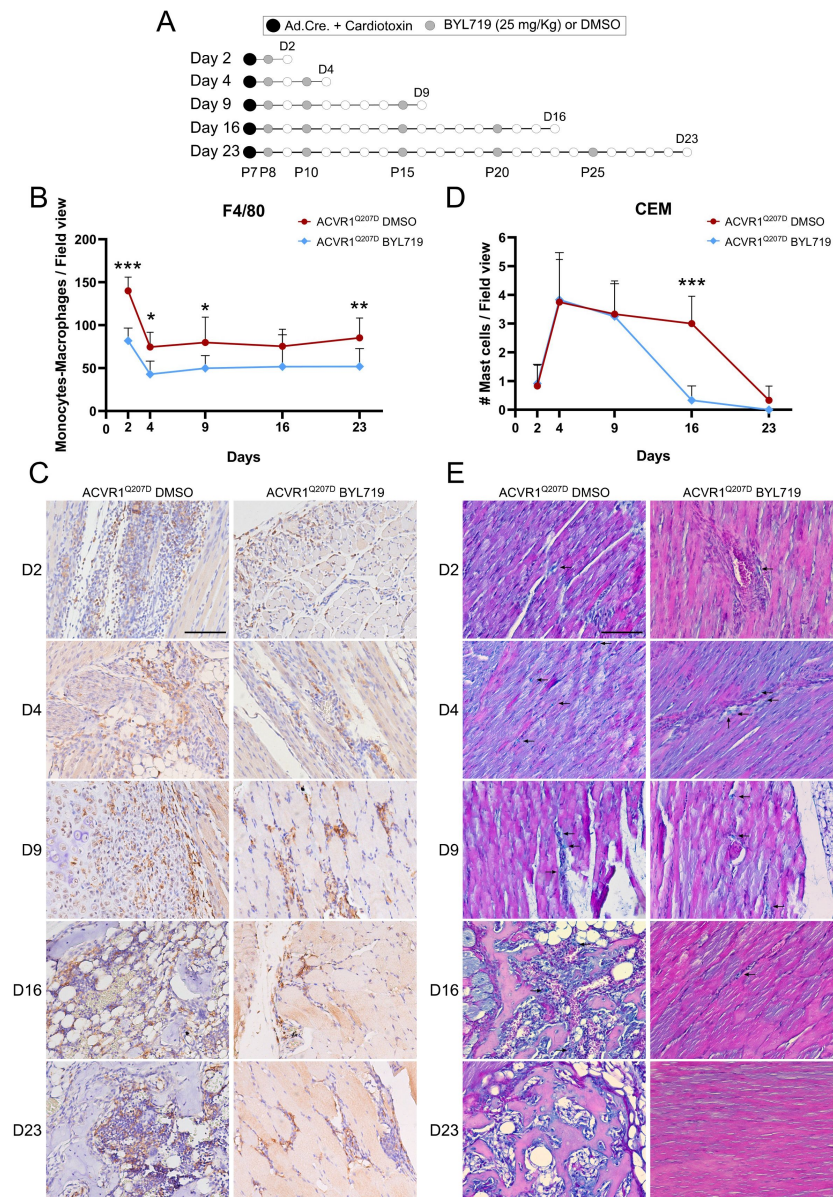


Figure 8.

(A) Heterotopic ossification was induced at P7 through the injection of Adenovirus-Cre (Ad.Cre) and cardiotoxin in the mice hindlimb. Either DMSO (vehicle) or BYL719 (25 mg/kg) were injected following the scheme indicated with grey dots, starting at P8 but with a different duration P9 (2 days post-HO-induction), P11 (4 days post-HO-induction), P16 (9 days post-HO-induction), P23 (16 days post-HO-induction), and P30 (23 days post-HO-induction). (B) The quantification of F4/80-positive monocytes/macrophages per field view from immunohistochemistry staining, with 20x amplification as shown in the representative images depicted in [Figure 8C](#). Five images were randomly acquired per mice, from 2 mice per group, for a total of 10 quantified images per group. F4/80 positive cells were detected as cells with dark brown staining. Data are shown as mean $\pm$ SD. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, two-way ANOVA with Sidak's multiple comparisons test, comparing both groups on each day ($n = 10$). (C) Representative images from the quantified immunohistochemistry staining for F4/80 to detect monocytes/macrophages. Representative images are shown for each final time-point at day 2, 4, 9, 16, and 23 post-HO-induction. All images were acquired at 20x (scale bar 100 μ m). (D) Quantification of CEM-positive mast cells per field view from CEM staining, with 20x amplification as shown in the representative images depicted in [Figure 8E](#). Three images were obtained per mice, from 4 mice per group, for a total of 12 quantified images per group. Mast cells were detected as cells with bright blue staining. Data are shown as mean $\pm$ SD. *** $P < 0.001$, two-way ANOVA with Sidak's multiple comparisons test, comparing both groups in each day ($n = 12$). (E) C.E.M. staining was performed to detect mast cells, highlighted with black arrows. Representative images are shown for each final time-point at day 2, 4, 9, 16, and 23 post-HO-induction. All the images were obtained at 20x magnification, with a representative scale bar at 100 μ m.

competitor inhibitor for none of the TGF- β receptor kinases. Finally, the administration of BYL719 prevented an exacerbated inflammatory response *in vivo*, possibly due to specific effects observed on immune cell populations.

While FAPs play a key role in bone formation in different types of HO, inflammation is also closely associated with all types of HO (Barruet et al., 2018; de Ruiter et al., 2021; Matsuo et al., 2019). For instance, the depletion of mast cells and macrophages has been shown to reduce bone volume in FOP mouse models (Convente et al., 2018). Moreover, Activin A, TGF- β and other cytokines secreted by FAPs, monocytes, macrophages and mast cells are essential for FOP and non-genetic HO (Alessi Wolken et al., 2017; Hattell et al., 2015; Lees-Shepard et al., 2018; Patel et al., 2022; Sorkin et al., 2020; Upadhyay et al., 2017). In addition, ACVR1 antibodies can activate ACVR1^{R206H} even in the absence of Activin A, but muscular trauma is still required to induce HO (Aykul et al., 2022; Lees-Shepard et al., 2022), indicating the involvement of additional immune factors. Our study finds that BYL719, a PI3K α inhibitor, reduces the expression of certain pro-inflammatory cytokines, effectively blocks the proliferation of monocytes, macrophages and mast cells and reduces the migratory potential of monocytes. This likely contributes to a decreased number of monocytes and macrophages at injury sites and throughout the *in vivo* ossification process. It is known that excessive pro-inflammatory cytokine expression, including activin A, by monocytes and macrophages is observed in all types of HO and is induced by ACVR1^{R206H} (Matsuo et al., 2021). Non-canonical signaling pathways (mTOR, p38, TAK1, NF- κ B) are also affected by ACVR1^{R206H} in immune cell types (Barruet et al., 2018; Hwang et al., 2022). M1-type macrophages initiate acute inflammatory responses but later transition to M2-type macrophages associated with anti-inflammatory and reparative functions. Therefore, modulating macrophage phenotype towards regeneration may dampen HO (Sorkin et al., 2020). BYL719 reduces monocyte/macrophage numbers, migration, and pro-inflammatory cytokine expression, potentially altering their polarization. It also reduces activin A expression and promotes a shift towards the M2 phenotype. BYL719 can help to reduce the hyper-inflammatory state in HO, inhibit FAP expansion, and favor myogenic regeneration of muscle tissue (Stanley et al., 2022).

Our observation that the delayed administration of BYL719 is still effective has pathophysiological and therapeutic derivatives. Muscle injury in mice induces sequential changes in HO progression: early immune cell infiltration (days 1-3), a fibroproliferative and late inflammatory phase, with an expansion of FAPs and a spectrum of monocytes, macrophages and mast cells as highly secretory cells (days 3-7), followed by chondrogenesis (day 7-14) and osteogenesis (days 14-23) (Hwang et al., 2022). Starting BYL719 administration 3 days post-injury is still fully therapeutically effective and even partially at 7 days post-injury. In addition, whereas untreated mice developed cartilage and bony lesions by days 9 and 16 post-injury, respectively, BYL719-treated animals fully regenerated muscle by day 16 post-injury. Altogether, this evidence suggests that the major effects of BYL719 occur between 3 and 16 days post-injury. We found that BYL719 completely blocks chondrogenesis in cultured hMSCs. In addition, in the histological analyses of these BYL719-treated mice that do not develop HO, we cannot find any sign of cartilage formation on either the 9th, 16th or 23rd days post-injury, which confirms that BYL719 is also able to block chondrogenesis in mice *in vivo*. This suggests that intervention with BYL719 over a temporal window after a HO flare might be sufficient to inhibit endochondral ossification. In addition, although surgical resection is not recommended in FOP patients and has a risk of recurrence in resected non-genetic HO, it could be envisaged using as preventive therapy in subjects undergoing surgeries to remove ectopic bone.

Treatment options for HO and FOP are currently limited, primarily consisting of anti-inflammatories such as corticosteroids and NSAIDs. Promising pharmacological treatments have progressed to II/III clinical trials. Palovarotene, a retinoic acid receptor- γ agonist, has been approved in Canada and U.S.A. for patients over 10 years old (NCT03312634). However, a previous clinical trial with palovarotene was paused for children due to concerns about premature growth plate closure, which is debated in FOP mice (Chakkalakal et al., 2016; Rosen et al., 2018).

(Chakkalakal et al., 2016 [↗](#); Rosen et al., 2018 [↗](#)). Antibodies targeting activin A (Garetosmab) showed effectiveness, but serious adverse effects led to a temporary hold on the trial (NCT03188666). The outcome of a clinical trial with Rapamycin has not been publicly disclosed and a case report indicated limited benefits in classical FOP patients (UMIN000028429) (Kaplan et al., 2018 [↗](#)). BYL719 (Alpelisib/Piqray®) was recently licensed in combination with hormone therapy for a PIK3CA-mutated breast cancer (Cardoso et al., 2020 [↗](#)). BYL719 has been shown to be also clinically effective in adult patients with PIK3CA-related overgrowth syndrome (PROS) and children under 1 year of age (Madsen and Semple, 2022 [↗](#); Morin et al., 2022 [↗](#)). In both scenarios, BYL719 had no major adverse effect when patients were treated for more than one year (in adults at a dose of 300mg daily for oncological or PROS purposes and at a dose of 25 mg daily in infants with PROS) (Morin et al., 2022 [↗](#)). Thus, implementing BYL719 for treatment of HO, at least during a restricted temporal window (i.e. surgery to remove ectopic bone in HO, or during flare-ups in FOP individuals), might be a valid therapeutic option for FOP patients.

Materials and Methods

Murine bone marrow mesenchymal stem cells isolation

Murine bone marrow-derived mesenchymal stem cells (BM-MSCs) were isolated from 6-8 weeks-old C57BL6/J mice for the floxed p110 α allele (Beatriz Gámez et al., 2016 [↗](#)) as previously described (Soleimani and Nadri, 2009 [↗](#)). Mice were euthanized and femurs were dissected and stored in DMEM with 100 U/ml penicillin/streptomycin (P/S). Soft tissues were cleaned and femur ends were cut under sterile conditions. Bone marrow was flushed with media using a 27-gauge needle. The resulting cell suspension was filtered through a 70- μ m cell strainer and seeded in a 100-mm cell culture dish. Non-adherent cells were discarded after 3 hours. Media was slowly replaced every 12 hours for up to 72 hours. Then, media was replaced every 2 days until the culture reached 70% confluence. Then, cells were lifted by incubation with 0.25% trypsin/0.02% EDTA for 5 minutes at room temperature. Lifted cells were cultured and expanded.

Production of human mesenchymal stem cells (hMSCs) expressing ACVR1

For chondrogenic differentiation and RNA sequencing, bone marrow-derived human MSCs were transduced by lentiviral delivery encoding ACVR1^{WT} or ACVR1^{R206H} (Van Dinther et al., 2010 [↗](#)). For NanoBRET target engagement kinase assays, were transduced with ACVR1^{WT}-nanoluc or ACVR1^{R206H}-nanoluc. The nanoluc fusion inserts (Promega, NV2341/NV2381) were subcloned into a PLV-CMV-IRES vector. First, an eGFP was amplified with PCR containing PstI, SalI and XbaI restriction sites and subcloned in the PLV using PstI and XbaI. Next, the ACVR1-nanoluc inserts were subcloned in the PLV by restriction of PstI and SalI, exchanging the eGFP. Subsequently, lentiviral particles were produced using HEK293t cells and cells were transduced.

Cell culture

Murine BM-MSCs were cultured in DMEM supplemented with 10% FBS, 2 mM glutamine, 1 mM sodium pyruvate, and 100 U/ml P/S and incubated at 37°C with 5% CO₂. The human monocyte cell line THP1 was cultured in RPMI-1640 supplemented with 10% FBS, 2 mM glutamine, 1 mM sodium pyruvate, 100 U/ml P/S, 0.1mM NEAA, 10mM HEPES and 50 μ M 2-mercaptoethanol. Mouse macrophage cell line RAW264.7 was cultured in DMEM supplemented with 10% FBS, 2 mM glutamine, 1 mM sodium pyruvate and 100 U/ml P/S. The human mast cell line HMC1 was cultured in IMDM supplemented with 10% FBS, 2 mM glutamine, 1 mM sodium pyruvate and 100 U/ml P/S. The hMSCs were cultured in α MEM supplemented with 10% FBS, 0.1 mM ascorbic acid, 1 ng/ml human bFGF and 100 U/ml P/S (Growth Medium; GM). C2C12 cells were cultured in DMEM

supplemented with 20% FBS, 2 mM glutamine, 1 mM sodium pyruvate, and 100 U/ml P/S and incubated at 37°C with 5% CO₂. For C2C12 differentiation, cells were cultured for seven days in the same media with 5% horse serum.

Proliferation and chemotactic migration assays

For proliferation assays, cells were seeded in 12-well culture plates in correspondent culture medium with 2 nM BMP6 and/or 2 μM or 10 μM BYL719 (Chemietek). Photos of several fields of each well were taken at 24, 48, 72 and 144 hours using a Leica DM-IRB inverted microscope. Cell counting was performed through ImageJ software.

The chemotactic migration assay of THP1 was performed through the InnoCyte Monocyte Migration Assay (Merck Millipore). Cells were serum-starved for 3h prior to the experiment and, when needed, treated with 2 μM BYL719 for 30 minutes before the experiment started. Then, 10⁵ cells/100 μL were added to each upper chamber. Lower chambers were charged with 10% FBS-supplemented RPMI-1640 (positive control), FBS-depleted RPMI-1640 (negative control) or 2 nM BMP6 (R&D), respectively. Cells migrated for 2h and were then stained with calcein-AM and pipetted into a 96-well conical bottom plate (Sigma-Aldrich). A plate reader was used to measure fluorescence from the top of the plate at an excitation wavelength of 485 nm and an emission wavelength of 520 nm. Unless otherwise stated, BYL719 at 2 or 10 μM (Chemietek) and 2 nM BMP6 (R&D) was used for 6 days (proliferation) or 4 hours (migration) in HMC1, THP1 and RAW264.7 cells.

Retroviral transduction and gene expression analysis

BM-MSCs were infected with mock virus (pMSCV) or with viruses expressing wild type *Acvr1* (WT), *Acvr1*^{R206H} (RH) and/or pMSCV-Cre. Plasmids with WT and RH *Acvr1* forms were kindly provided by Dr. Petra Seemann and were subcloned into pMSCV vector. *Acvr1* and p110α expression levels were analyzed by qRT-PCR. Cells were treated with BYL719 (2 μM) and/or activin A (2 nM) for 48 hours in complete media without FBS. THP1, RAW264.7, and HMC-1 cell lines were treated in the media indicated for each cell line with BYL719 (2 μM) and/or BMP6 (2 nM) for 48 hours.

Total RNA for RT-PCR was extracted from all cellular models using TRIpure reagent. At least 2 μg of purified RNA was reverse-transcribed using the High-Capacity cDNA Reverse Transcription Kit (Applied Biosystems). Quantitative PCRs were carried out on ABI Prism 7900 HT Fast Real-Time PCR System with Taqman 5'-nuclease probe method and SensiFAST Probe Hi-ROX Mix. All transcripts were normalized using *Tbp* as an endogenous control.

NanoBRET target engagement assays

COS-1 cells were cultured in DMEM (Gibco, 11965092) supplemented with 10% FBS and 100 U/ml P/S. These cells were transfected in 70% confluent 6-wells TC-treated plates with 2 μg TGFβ receptor-Nanoluciferase constructs (constructs were provided by Promega and Promega R&D) using a 1 μg DNA: 2 μl PEI (1 mg/ml) ratio. The transfected cells were reseeded in white 384-wells TC-treated assay plates one day before the nanoBRET readout in DMEM supplemented with 1% FBS in a quantity of 15*10³ cells in 40 μl per well. Two hours before measurement, the dedicated tracer (Promega) and the test compounds were incubated to the cells at 37°C. Immediately after intracellular TE Nano-Glo Substrate/Inhibitor (Promega, N2160) addition, the wells were measured by 450-80BP and 620-10BP filters using the ClarioSTAR (BMG Labtech). Subsequent nanoBRET ratios were measured by the formula: acceptor emission (620-10 nm)/donor emission (450-80 nm) * 1000 (milliBRET units, mBU).

<i>TGFR2-nluc</i>	Promega R&D	K-11 (0.32 μ M)	N2652	BYL719 (1 and 10 μ M) ML347 (10 μ M)
<i>BMPR2-nluc</i>	Promega R&D	PBI7394 (0.5 μ M)	Promega R&D	BYL719 (1 and 10 μ M) ML347 (10 μ M)

Chondrogenic progenitor differentiation

The hMSCs were cultured in α MEM supplemented with 10% FBS, 1 ng/ml human bFGF and 100 U/ml P/S (Growth Medium; GM). Chondrogenic differentiation medium (DM) consisted of DMEM/F12 supplemented with 1% Insulin-Transferrin-Selenium, 170 μ M Ascorbic Acid, 0.1 μ M Dexamethasone, 350 μ M L-proline, 1 mM Sodium Pyruvate, 0.15% glucose, 1%FBS and 100 U/ml P/S. During differentiation, the medium was refreshed twice a week for the duration of the experiment. For the Alcian Blue experiments, micromass cultures were made by carefully seeding 3×10^5 cells per 10 μ l droplets in a 24-wells plate and incubated for 2 hours at 37°C prior to GM addition. Chondrogenic progenitor differentiation was started 1 day after micromass seeding by the addition of DM. Dependent on the condition, we supplemented the DM with TGF- β 1 (10 ng/ml) or Activin A (100 ng/ml), and treated with BYL719 (10 μ M) or DMSO vehicle. For the ALP staining, the hMSC-ACVR1 lines were seeded at 2×10^4 cells per well in 48-wells plates and differentiated for 7 and 11 days.

Alkaline phosphatase (ALP) staining, ALP activity assay and Alcian Blue staining

For the ALP staining, the cells were washed with PBS, fixated using 3.7% formalin for 5 minutes at RT, washed twice with PBS and stained in ALP solution containing 2 mg Naphthol AS-MX, 6 mg Fast Blue, 5 ml 0.2M Tris (pH 8.9), 100 μ l MgSO_4 and dH_2O up to 10 ml. Images were acquired using the Leica DMI8 with a 10x magnification. To measure ALP activity, ALP lysates were obtained by washing twice with PBS, freezing at -80°C for 1 hour, and lysing on ice for 1 hour using 100 μ l ALP buffer (100 μ M MgCl_2 , 10 μ M ZnCl_2 , 10 mM Glycine (pH 10.5)) plus 0.1% Triton X-100 per well. ALP activity was quantified by adding 20 μ l lysate and 80 μ l 6 mM PNPP in ALP buffer to a clear 96-well plate incubated at RT until the samples turned yellow. Next, the samples were measured at an absorbance of 405 nm.

The micromass pellets were stained with Alcian Blue as described before (Sánchez-Duffhues et al., 2019). Images were acquired using the Leica DMI8 with 5x magnification. Densitometry analysis of the whole well (from total plate scan) was performed using ImageJ (v1.53t).

Bulk RNA sequencing and analyses

The hMSCs-ACVR1^{WT} and hMSCs-ACVR1^{R206H} were seeded at 100% confluency and serum starved overnight with DMEM without supplements. Afterwards, the cells were pre-treated with or without 1 μ M of BYL719 in starvation medium for 30 minutes and stimulated with or without 50 ng/ml Activin A or 50 ng/ml BMP6 for 1 hour. RNA was isolated using the ReliaPrep RNA Miniprep Systems (Promega) and sequenced using the Illumina NovaSeq 6000 (Illumina) platform. The reads were mapped using the Hisat2 (v2.0.5) package and the total and normalized counts were measured using featureCounts (v1.5.0-p3). Differential expression analysis was performed using the DESeq2 R package (v1.20.0), in which the p-values were adjusted following the Benjamini and Hochberg approach. Genes with an adjusted p-value < 0,05 were considered differentially expressed. Subsequent Gene Ontology analysis was performed using the clusterProfiler R package and the GO terms were considered significant if the p-values < 0,05. Gene Set Enrichment Analysis was performed using the GSEA analysis tool ([gsea-msigdb.org](https://www.broadinstitute.org/gsea/)). Heatmaps were generated using heatmap2 function within the R gplots package in Galaxy (usegalaxy.eu). The datasets are publicly available (see GSE237512).

Western blot

Cells were lysed using RIPA lysis buffer and quantified using the Pierce BCA Protein assay (Thermo) following standard protocols. The protein samples were loaded using β -ME containing sample buffer and run using SDS-PAGE and transferred to PVDF membranes. Primary antibodies against SMAD1 (Cell Signaling, 6944), p-SMAD1/5 (Cell Signaling, 9516), and vinculin (Sigma, V9131) were used at 1:1000 dilution in 5% BSA in TBST. Binding was detected with HRP-conjugated secondary antibodies and visualized by Brightfield ECL (Thomas Scientific) on the ChemiDoc (BioRad).

In vitro recombinant kinase activity assay

Fifty ng of recombinant ACVR1^{R206H} (ab167922, Abcam) was diluted in 30 μ l of 25mM Tris/HCl (pH7.5), 10mM MgCl₂, 10mM MnCl₂ and 2mM DTT. The reaction was started by the addition of 5 μ M ATP-Mg, 0.05 μ Ci/ μ l ATP [γ -³²P] and 0.5mg/ml of dephosphorylated casein (Sigma) and performed for 30 min at 30°C. The reaction was stopped and samples were processed by SDS-PAGE. After drying the gel, phosphorylation was determined by autoradiography.

Heterotopic ossification mouse model

To study heterotopic ossification *in vivo*, we used the Cre-inducible constitutively active ACVR1^{Q207D} (CAG-Z-EGFP-caALK2) mouse model as previously described (Fukuda et al., 2006; Shimono et al., 2011; Yu et al., 2008). Mice ACVR1Q207D^{fl/fl} p110 α ^{fl/fl} were obtained by crossing the detailed homozygous ACVR1Q207D^{fl/fl} with the p110 α ^{fl/fl} homozygous mutant mice carrying loxP sites flanking exons 18 and 19 of the p110 α alleles (Beatriz Gámez et al., 2016). Heterozygous mice were crossed until ACVR1Q207D^{fl/fl} p110 α ^{fl/fl} was obtained.

To induce heterotopic ossification in P7 ACVR1^{Q207D} mice, 1 $\times$ 10⁸ pfu of adenovirus-Cre (Ad-CMV-Cre, Viral Vector Production Unit, UAB) and 0.3 μ g of cardiotoxin in 10 μ l 0.9% NaCl volume were injected into the left hindlimb. Control groups had the same procedure without Ad-Cre in the injection. On P8, mice started receiving either placebo (intraperitoneal administration (i.p.) of DMSO) or BYL719 (i.p. of 25 mg/kg), both diluted in 0.5% carboxymethylcellulose sodium. Mice were housed under controlled conditions (12-hour light/12-hour dark cycle, 21°C, 55% humidity) and fed *ad libitum* with water and a 14% protein diet (Teklad2014, Envigo). Mice were regularly weighted over the whole period. Heterotopic ossification induction was performed blinded to mouse and group identity. All procedures were approved by the Ethics Committee for Animal Experimentation of the Generalitat de Catalunya.

Micro-computed tomography analysis

The caudal half of mice was collected and fixed in 4% paraformaldehyde (PFA) for 48 hours at 4°C. Samples were conserved in PBS and high-resolution images were acquired using a computerized microtomography imaging system (Skyscan 1076, Bruker microCT), in accordance with the recommendations of the American Society of Bone and Mineral Research (ASBMR). Samples were scanned in air at 50LkV and 200L μ A with an exposure time of 800Ljms, using a 1Ljmm aluminum filter and an isotropic voxel size of 9L μ m. TwoLjdimensional images were acquired every 1° for 180° rotation and subsequently reconstructed, analyzed for bone parameters, and visualized by NRecon v1.6, CT-Analyser v1.13 and CTVox v3.3 programs (Bruker), respectively. For heterotopic ossification, manual VOIs comprising heterotopic ossifications were employed and a binary threshold was established at 25-255.

Histology

Whole legs were fixed in 4% PFA for 48 hours at 4°C, decalcified in 16% EDTA pH 7.4 for 6 weeks, and embedded in paraffin. Five μm sections were cut and stained with Fast Green/Safranin O, or Masson's Trichrome. Images were obtained with brightfield Eclipse E800 (Nikon). To detect monocyte/macrophage cells, paraffined slides were immunostained with Anti-F4/80 antibody [SP115] (ab111101). Briefly, after deparaffination and rehydration, an antigen retrieval step with citrate buffer at pH6 in a decloaking chamber was applied to the slides. Slides were cooled down and rinsed with 1x TBS and endogenous peroxidases were blocked with 70% methanol, 28% distilled water, and 2% hydrogen peroxide for 5 minutes at room temperature. After a blocking step, F4/80 primary antibody incubation at concentration 1:100 overnight at 4 °C was applied. Envision Dual Link conjugated with HRP was applied for 40 minutes at room temperature, and detection was performed with diaminobenzidine (DAB) for 3 minutes. Nuclear staining with Lilly's hematoxylin was applied to the slides for 30 seconds. Images of stained slides were obtained with brightfield Eclipse E800 (Nikon). Images were randomly acquired with 20x magnification from the region of interest (injected muscle, with or without visible heterotopic ossification). Five images were obtained per mice, from 2 mice per group, for a total of 10 quantified images per group. F4/80 positive cells were detected as cells with dark brown staining. To detect mast cells, paraffined slides were stained with C.E.M. staining using the Eosinophil-Mast cell staining kit (ab150665, Abcam). The recommended staining procedure was applied to the deparaffinized sections. Images of stained slides were obtained with brightfield Eclipse E800 (Nikon). Images were randomly acquired with 20x magnification from the region of interest (injected muscle, with or without visible heterotopic ossification). Three images were obtained per mice, from 4 mice per group, for a total of 12 quantified images per group. Mast cells were detected as cells with bright blue staining.

Statistical analysis

Unless stated otherwise in each figure legend, the results were expressed as mean $\pm$ SD. The median was shown in heterotopic ossification datasets where non-parametric tests were applied. Each figure legend has detailed information explaining the statistical test used to compare between groups, the compared groups, the explanation of the symbols used to show significance, usually * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, and the sample size of the experiment. Heterotopic ossification, evaluation, and quantification was performed blinded to mouse and group identity. Microscopy and histology representative images were selected from the total quantified images. Microscopy and histology quantified fields were randomly acquired from the plate or the region of interest. Statistical tests were performed on GraphPad Prism 9.5.

Acknowledgements

We thank E. Adanero, E. Castaño for technical assistance. We acknowledge the support from Promega R&D. Alexandre Deber is a recipient of a F.P.I. fellowship from the Ministry of Science and Innovation (MCIN/AEI/ 10.13039/501100011033). Carolina Pimenta-Lopes is the recipient of a F.P.U. fellowship from the Spanish Ministry of Education. This research was supported by grants PID2022-141212OA-I00, PDC2021-121776-I00 and PID2020-117278GB-I00 from MCIN/AEI/10.13039/501100011033, co-funded by FEDER 'Una manera de hacer Europa' or 'NextGenerationEU'/PRTR; a grant 202038-30 from La Marató de TV3 and grants from IFOPA (ACT for FOP) and FOP Italia. MW and MJG are sponsored by the Netherlands Cardiovascular Research Initiative (the Dutch Heart Foundation, Dutch Federation of University Medical Centers, the Netherlands Organization for Health Research and Development, and the Royal Netherlands

Academy of Sciences), PHAEDRA-IMPACT (DCVA) and DOLPHIN-GENESIS (CVON). GSD is also sponsored by the Spanish Ministry of Science through the Ramón y Cajal grant RYC2021-030866-I and the BHF-DZHK-DHF, 2022/23 award PROMETHEUS.

Author contributions

J.A.V., A.D., M.W., M.J.G., J.L.R., G.S-D. and F.V. designed the study, and analyzed and interpreted the results. J.A.V, A.D., M.W. and C.P-L. performed the experiments. J.A.V., A.D., G.S-D. and F.V. wrote the manuscript. All authors contributed to the manuscript. G.S-D. and F.V. supervised the project.

Competing interests

The authors declare no conflicts of interest.

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Reviewer #1 (Public Review):

Summary:

In the present study, the authors examined the possibility of using phosphatidyl-inositol kinase 3-kinase alpha (PI3K α) inhibitors for heterotopic ossification (HO) in fibrodysplasia ossificans progressiva (FOP). Administration of BYL719, a chemical inhibitor of PI3K α , prevented HO in a mouse model of FOP that expressed a mutated ACVR1 receptor. Genetic ablation of PI3K α (p110a) also suppressed HO in mice. BYL719 blocked osteochondroprogenitor specification and reduced inflammatory responses, such as pro-inflammatory cytokine expression and migration/proliferation of immune cells. The authors claimed that inhibition of PI3K α is a safe and effective therapeutic strategy for HO.

This is a revision of the original manuscript by Valer *et al.* The authors performed new experiments and added those data to the manuscript to respond to this reviewer's comments and questions.

Strengths:

Now it became clear that BYL719 inhibited the multiple signaling pathways in multiple types of cells.

Weaknesses:

However, it was not clear the critical role of PI3K in the inhibition of HO by this compound.

<https://doi.org/10.7554/eLife.91779.2.sa1>

Reviewer #2 (Public Review):**Summary:**

The authors in this study previously reported that BYL719, an inhibitor of PI3K α , suppressed heterotopic ossification in mice model of a human genetic disease, fibrodysplasia ossificans progressive, which is caused by the activation of mutant ACVR1/R206H by Activin A. The aim of this study is to identify the mechanism of BYL719 for the inhibition of heterotopic ossification. They found that BYL719 suppressed heterotopic ossification in two ways: one is to inhibit the specification of precursor cells for chondrogenic and osteogenic differentiation and the other is to suppress the activation of inflammatory cells.

Strengths:

This study is based on authors' previous reports and the experimental procedures including the animal model are established. In addition, to confirm the role of PI3K α , authors used the conditional knock-out mice of the subunit of PI3K α . They clearly demonstrated the evidence indicating that the targets of PI3K α are not members of TGFBR by a newly established experimental method.

Weaknesses:

Overall, the presented data were closely related to those previously published by authors' group or others and there were very few new findings. The molecular mechanisms through which BYL719 inhibits HO remain unclear, even in the revised manuscript.

Heterotopic ossification in the mice model was not stable and inappropriate for the scientific evaluation.

The method for chondrogenic differentiation was not appropriate, and the scientific evidence of successful differentiation was lacking.

The design of the gene expression profile comparison was not appropriate and failed to obtain the data for the main aim of this study.

The experiments of inflammatory cells were performed in cell lines without ACVR1/R206H mutation, and therefore the obtained data were not precisely related to the inflammation in FOP.

<https://doi.org/10.7554/eLife.91779.2.sa0>

Author response:

The following is the authors' response to the original reviews.

Our revised version of the manuscript addresses all the comments and suggestions raised, as clarified in our point-by-point answer to the reviewers. We have performed additional experiments regarding the effects on proliferation and differentiation of additional cell types in the muscle, such as myogenic and mesenchymal progenitors as well as chondrogenesis in parental hMSCs that did not express exogenous ACVR1. Moreover, as suggested by reviewer #2, we performed all the chondrogenic experiments with addition of TGF β in the differentiation media and analyzed chondrogenesis by both Alcian blue staining and qPCR analysis of gene markers (Sox9, Acan, Col2a1 and Mmp3). We also extended our RNA-seq analysis and included new data using both hMSCs expression wild type or R206H ACVR1 receptor, with or without different ACVR1 ligands (BMP6 and Activin A) and treated or not with the inhibitor BYL719. The new data suggests that BYL719 is able to inhibit the expression of genes involved in ossification and osteoblast differentiation irrespective of the presence of the mutation. We also discuss the effect of BYL719 in mTOR signaling and addressed all the minor comments suggested by both reviewers.

We addressed the specific comments of the reviewers as follows:

Reviewer # 1:

Specific points:

Point #1 and #2. The authors showed that BYL719 inhibited HO in FOP model mice. Did they have HO not only in the muscle but also in the bone marrow? The progenitor cells of chondrocytes and osteoblasts may differ between the muscle and bone marrow. The authors should examine the effects of BYL719 on some other types of cells in the muscle, such as myoblasts and fibro-adipogenic cells, in addition to the bone marrow-derived MSCs. Furthermore, it was unclear whether they were human or murine MSCs in the text.

The inhibitory effect of BYL719 on HO in FOP mice was clear, but the molecular mechanisms or target cells were still unclear because BYL719 affected multiple types of cells and molecules. The authors are encouraged to show clearer mechanisms and target cells' critical inhibition of HO. Again, this reviewer believes that in vivo and in vitro experiments using muscle and bone marrow and cells prepared from them should provide additional critical information.

As detailed in the introduction, it is known that Heterotopic Ossification develops in the skeletal muscle and connective tissues. Consistent with the current knowledge of the field, none of the mice showed HO in the bone marrow. Additionally, since activation of the mutant allele is achieved by injection of CRE-expressing adenovirus and cardiotoxin in the muscle hindlimb, it is unlikely that mesenchymal progenitors in the bone marrow would be strongly affected. Interestingly, single-cell RNA sequencing from multiple mouse tissues identified a very strong transcriptional similarity between FAPs and non-muscle mesenchymal progenitors (PMID: 37599828). As suggested, we examined the effects of BYL719 in proliferation and differentiation in additional cell types such as muscle progenitors. In this new version of the manuscript, we show that BYL719 reduces the proliferation of muscle and mesenchymal progenitors while it blocks myoblast differentiation in vitro (Figure 7, Figure Supplement 1). MSCs were murine on those experiments shown in Figure 3; whereas assays shown in Figures 5 and 6 were of human origin. We have further clarified this in the respective Figure legends.

All the data generated strongly suggests that there is not a single mechanism supporting all the effects of BYL719 in HO. Instead, BYL719 affects multiple cell types involved in efficient HO (e.g. reduction in proliferation and osteochondrogenic specification of mesenchymal precursors (MPs), reduction on proliferation, migration, and inflammatory gene expression on monocytes, etc.). Interestingly, our data suggests that BYL719 is able to inhibit these effects

on MPs and monocytes irrespective of the presence of the ACVR1-R206H mutation (Figures 5, 6 and 7). Additionally, there are several signaling mechanisms affected. BYL719 reduces SMAD1/5, p38, AKT and mTOR signaling in parental MPs or with mutations in ACVR1 (Figure 3 and our previous publication PMID: 31373426), being all these pathways required for efficient osteochondrogenic specification of MPs. We consider that the different detailed mechanisms by which BYL719 inhibits osteochondrogenic specification enhances the robustness of the findings in this study.

Point #3. In FOP model mice, ACVR1 was mutated as Q207D. However, R206H was used in in vitro experiments. Do they have the same characteristics? This reviewer would like to recommend examining the effect of BYL719 on wild-type ACVR1, R206H, and Q207D simultaneously in each experiment.

We already performed these experiments, assaying in parallel ACVR1-WT, ACVR1-Q207D and ACVR1-R206H, in the transcriptional responses of MPs in our previous work (PMID: 31373426). Both mutations had similar responses, being ACVR1-Q207D stronger than ACVR1-R206H, as it has been shown in vivo in mouse models of HO (PMID: 34633114). In any case, BYL719 inhibits these transcriptional responses induced by both mutant alleles.

Point #4. Figure 5: What was the effect of BYL719 on the differentiation of parental cells that did not express exogenous ACVR1?

We performed new assays of chondrogenic differentiation of hMSCs that are shown in the new Figure 5. BYL719 inhibits chondrogenic differentiation of parental hMSCs and also inhibits chondrogenic specification irrespective of the expression of either wild type or mutant ACVR1.

Point #5. Figure 6: In this experiment, gene expression was examined in pretreated MSCs-ALK2 (ACVR1?) R206H with and without BYL719. It was clear whether suppression of gene expression by BYL719 was specifically caused in cells expressing R206H. What were the effects of BYL719 on parental cells that did not express exogenous ACVR1?

To be consistent, we relabeled ALK2 to ACVR1 in the figure. We expanded the conditions analyzed in the RNA-sequencing. We included conditions where we activate ACVR1 (either WT or R206H) with their known physiological ligand BMP6. In both, human MSCs expressing ACVR1-R206H and human MSCs expressing Wild Type ACVR1, we observed a downregulation of differentially expressed genes upon addition of BYL719, irrespective of ligand (BMP6 or Activin A) or receptor (RH or WT) (added new Figure 6: B and C).

Point #6. Figure 7: BYL719 suppressed cell proliferation of all cells examined partially at 2 uM and almost completely at 10 uM, respectively. There is a possibility that BYL719 inhibits HO by inhibiting osteochondrogenitor proliferation. The authors are encouraged to show data on the effect of BYL719 on the proliferation of other types of cells, such as myoblasts, fibro-adipogenic cells, or bone marrow cells.

We examined the effects of BYL719 in proliferation in additional cell types such as muscle and mesenchymal progenitors. BYL719 slightly reduced the proliferation of myoblasts and mesenchymal cells in vitro (Figure 7, Figure Supplement 1). However, the reduction in the proliferation in myoblasts or MPs did not reach the extent to that observed in monocytes or macrophages (Figure 7).

Point #7. Figure 8: How was the effect of BYL719 on muscle regeneration in wild-type? It was reported that mTOR signaling is important in HO in FOP. The authors are encouraged to show the effect of BYL719 on mTOR signaling.

Muscle regeneration in wild-type mice has also been shown in our previous results PMID: 31373426. In addition, we included images of the muscle regeneration after 23 days of treatment with BYL719 in mice ACVR1Q207D with or without PI3K α deletion after induction of HO in the new Figure 2, Figure Supplement 2. These mice showed full muscle regeneration or small calcifications surrounded by muscle at most. The effects of PI3K α inhibitors, either BYL719 or A66, on mTOR signaling had been previously shown by our group (PMID: 31373426). Both inhibitors strongly reduced signaling of mTOR, visualized by activation of p70 S6-kinase, a surrogate marker of mTOR activity.

Minor points:

(9) SMAD 1/5 should be SMAD1/5.

(10) The source of human MSCs should be indicated in the text.

(11) ALK2 should be ACVR1 in Figure 6A.

(12) The protein levels of each receptor should be examined in Fig. 4.

We introduced the suggested changes in the manuscript and Figure 6 and indicated the source of human MSCs in Materials and Methods. We also examined the levels of each receptor that are shown in the new Figure 4, Figure Supplement 1.

Reviewer # 2:

Specific points:

Point #1. Because the involvement of PI3K in HO of FOP, was already reported by authors' group and also others (Hino et al, Clin Invest, 2017), the main purpose of this study was to disclose the mechanism of how PI3K was activated in FOP cells. In the published study (Hino et al, Clin Invest, 2017), PI3K was activated by the ENPP2-LPA-LPR cascade. Unfortunately, there were no new data for this important issue.

The main purpose of this study is to demonstrate that the pharmacological and genetic inhibition of PI3K α in HO progenitors at injury sites reduces HO in vivo, to extend the insights into the molecular and cellular mechanisms responsible for the therapeutic effect of PI3K inhibition, and to optimize the timing of the administration of BYL719. Class I PI3Ks are heterodimers of a p110 catalytic subunit in complex with a regulatory subunit. They engage in signaling downstream of tyrosine kinases, G protein-coupled receptors and monomeric small GTPases. Therefore, a plethora of growth factors, cytokines, inflammatory agents, hormones and additional external and internal stimuli are able to activate PI3K α (PMID: 31110302). In fact, TGF- β family members, including activin A, are able to activate PI3K and mediate some of their non-canonical responses (PMID: 19114990). Multiple factors with known increased expression in the ossifying niche in HO and FOP (e.g. activin A, TGF- β , inflammatory agents such as TNF α , IL6, IL3, etc.) are known activators of PI3K (PMID: 30429363). Interestingly, in our RNA-seq analysis in hMSCs we did not observe increased expression levels of Enpp2 when comparing wild type and R206H mutated cells treated with activin A.

Point #2. The HO formation of ACVR1/Q207D model mice in this study is extremely unstable (Figure 1B, DMSO). Even the bone volume of some red symbols, which indicate the presence of HO, is located on the base (0.00) line. I would examine carefully the credibility of the data. Also, it is well known that the molecular behavior of mice Acvr1/Q207D and human ACVR1/R206H was different.

We agree with the reviewer that induction of HO is variable between mice showing variations in penetrance and intensity of the ossifying lesions. This variability is a known common trend that appears in all the models of HO published so far (e.g. PMID: 28758906, PMID: 26333933). Accordingly, we did not exclude any animal that has been injected with CRE-expressing adenovirus plus cardiotoxin in the μ CT analysis. Regarding the behavior of mice *Acvr1/Q207D* and human *ACVR1/R206H*, it is well known that *Q207D* produces more robust and stronger responses in terms of signaling and formation of heterotopic ossification (PMID: 34633114). Therefore, reduction of HO by *BYL719* would be more stringent in the *Acvr1/Q207D* model.

*Point #3. The experimental design of Figure 5 experiments is confusing. Although the authors mentioned that the data in Figure 5A were taken seven days after chondrogenic induction, I am skeptical whether the chondrogenic induction was successful. Based on the description of Material and Methods, the authors did not include TGF β in their "Differentiation Medium", which is an essential growth factor to induce chondrogenic differentiation of human MSC. Why did the ALP activity increase after chondrogenic induction? The authors should demonstrate the evidence of successful chondrogenic induction by showing the expression of key chondrogenic genes such as SOX9, ACAN, or COL2A1. The data in Figure 5B-E are also confusing. The addition of Activin A showed no difference between *ACVR1/WT* and *ACVR1/R206H* cells, suggesting that these cells did not reproduce the situation of FOP.*

We performed new assays of chondrogenic differentiation of hMSCs that are shown in the new Figure 5. We included TGF β 1 in the differentiation medium and also included the parental cell line in the analysis. In addition of being a marker of osteoblast differentiation, alkaline phosphatase (ALPL) has also been shown to be induced during chondroblast differentiation in vitro (PMID: 19855136; PMID: 9457080; PMID: 18377198; PMID: 23388029). Moreover, expression data of SOX9, COL2A1, ACAN and MMP13 of cells after chondrogenic differentiation is included in the new Figure 5. Expression of some markers (e.g. ACAN) are increased by the expression of *ACVR1/R206H*, however, we did not observe significant differences in chondroblast differentiation gene expression between *ACVR1wt* and *ACVR1/R206H* expressing cells. In any case, *BYL719* could inhibit chondrogenic differentiation of parental hMSCs and also the chondrogenic specification irrespective of the expression of either wild type or mutant *ACVR1*.

*Point #4. The experimental design and data analyses of RNA-seq were inappropriate and insufficient, which is disappointing for the reviewer because this will be a key experiment in this study. Because the most important point is to identify the signal for PI3Ka induced by Activin A via *ACVR1/R206H*, they should also use hMSC-*ACVR1/WT* for this experiment. Because the authors clearly demonstrated that TGFBR were not targets of *BYL719*, they should compare the expression profiles between MSC-*ACVR1/WT* and MSC-*ACVR1/WT* with *BYL719* to identify the targets of *BYL719* unrelated to Activin A signal. Then the expression profiles of *ACVR1/R206H* cells treated with Activin A and Activin A plus *BYL719* were compared. Among down-regulated signals by *BYL719*, those found also in MSC-*ACVR1/WT* should be discarded. It is important to investigate whether the GO term of ossification or osteoblast differentiation is found also in MSC-*ACVR1/WT*. If it is so, the effect of *BYL719* is not specific for FOP cells.*

We extended our RNA sequencing analysis with additional experimental conditions and comparisons. In new Figure 6, we now compare hMSCs expressing wild type or *R206H* receptors, with or without *BYL719* inhibition, and with or without different ligand activations (BMP6 or Activin A) (New Figure 6A). New Figure 6B shows the Gene ontology analysis of the differentially expressed genes between cells expressing WT and RH receptors under control

conditions. We can observe that ossification (GO:0001503) and osteoblast differentiation (GO:0001649) were detected within the top 10 significantly differentially regulated biological processes between these conditions. Therefore, we analyzed these relevant identified GO terms in 5 different comparisons upon GO enrichment analysis (Figure 6C). In addition to the comparison between cells expressing WT and RH receptors under control conditions explained above, we also compared cells expressing WT or RH receptor, with different ACVR1 ligands (BMP6 and Activin A), and with or without BYL719 inhibitor. The addition of BYL719 resulted in a downregulation of the GO terms “ossification” and “osteoblast differentiation” (new Figure 6C). These results confirm the inhibitory effect of BYL719 on ossification and osteoblast differentiation biological processes, and inform that this inhibitory effect remains consistent upon BMP6 or Activin A ligand activation, and with ACVR1 WT and RH expression.

Point #5. The data in Figure 7 were not related to the aim of this study because cell lines used in these experiments did not have ACVR1/R206H mutations. It is not appropriate to extrapolate these data in the FOP situation.

We utilized immune cell lines where we could activate ACVR1 with their known physiological ligand BMP6. Mutated ACVR1 gains response to activin A in addition to maintaining the physiological response to BMP6 as the wild type form. Therefore, in these assays we interrogated in vitro, with addition of BMP6, the effects of BYL719 in the growth, migration and inflammatory gene expression upon conditions of activated ACVR1 receptor downstream signaling. We consider that understanding the effects of PI3Ka inhibition in the regulation of proliferation, migration and inflammatory cytokine expression in monocytes, macrophages and mast cells is essential to better define the potential outcome of BYL719 treatment for heterotopic ossifications.

Minor comments:

(1) The legends for Figure 1C were those for Figure 1D, and there were no descriptions for Figure 1C in the legends and methods section. The reviewer was unable to understand the meaning of BV/TV. What is TV?

(2) “However, in PI3Ka deficient mice ACVR1Q207D expression only led to minor ectopic calcifications that were already surrounded by fully regenerated muscle tissue on the 23rd day after injury (Figure 2D, Figure 2-Figure Supplement 1B)”: There were no histological data either Figure 2D, Figure 2-Figure Supplement 1B), which showed muscle tissues.

(3) “The overexpression of Acvr1R206H increased basal and activin dependent expression of canonical (Id1 and Sp7) and non-canonical (Ptgs2) BMP target genes (Figure 3C),”: There was no increase of Ptgs2 gene in basal level.

(4) Materials and Methods. Production of human fetal mesenchymal stem cells expressing ACVR1.: Is it derived from a fetus?

(5) Figure 6C: There was no description of the meaning of each column. What does AA mean and what is the number?

We introduced the missing information in the manuscript, Figure legends and material and methods section for points #1, 4 and 5. AA was Activin A, the number was the number of replicates. This has been detailed in the figure legend. We included images of the muscle regeneration after 23 days of treatment with BYL719 in mice after induction of HO in the new Figure 2, Figure Supplement 2 (point #2). We corrected the mistake in the manuscript refraining for suggesting increase of Ptgs2 gene expression by ACVR1-R206 at the basal level (Point #3).

<https://doi.org/10.7554/eLife.91779.2.sa3>